Quadratic Chirp and Fixed-Axis Core Collection from a Finite-Energy Airy Beam

Auto BeamScientist Project

Abstract

A quadratic phase changes transverse phase gradients without changing the source-plane intensity. We numerically test the registered prediction that a chirp of amount $c = -0.5$ raises the median fraction of power inside a fixed on-axis circular core for a finite-energy Airy source with scale 0.20 mm and apodization 0.10, propagated in free space from 0.04 to 0.08 m. In the executed 384×384 angular-spectrum calculation, the transformed and base median core fractions are 0.072402 and 0.062833, respectively, giving a ratio of 1.1528. A 576×576 reference calculation gives the same ratio to four decimal places. Mesh, window, axial-sampling, and independent paraxial Fresnel checks give ratios near 1.153; the zero-chirp identity control is validated. A measured $c = +0.5$ comparison gives a ratio below unity, about 0.876, but is a secondary comparison and does not establish a general chirp-sign law. Centroid-aligned collection is higher than fixed-axis collection in both arms, so the primary endpoint reflects both position and intensity distribution. The result supports the registered numerical prediction for this source and collection geometry, but does not isolate a unique mechanism or establish a distinction from the broader literature on chirped Airy beams.

1 Introduction

Finite-energy Airy beams are studied for their structured profiles and propagation behavior [2, 4]. A phase-only intervention preserves the input intensity but changes the transverse phase gradients. During propagation, those gradients affect diffraction and can change the power collected within a prescribed region. The relevant outcome depends on what is collected: peak intensity, power inside a fixed aperture, and power inside an aperture recentered on the beam are different quantities.

This work tests a model-authored, registered question: for a finite-energy Airy source with $x_0 = 0.20$ mm and apodization 0.10, does a quadratic chirp of amount -0.5 yield a transformed-to-base ratio of median fixed-axis core efficiency at least one over $z = 0.04$–0.08 m? The proposed physical explanation is that the chirp changes diffraction-mediated collection through its effect on phase gradients. The fixed-axis observable is also sensitive to beam displacement, however, and cannot by itself distinguish displacement from redistribution of intensity.

The calculation is scalar and numerical. We compare matched chirped and untransformed fields at equal input power, with the same aperture, wavelength, propagation settings, and axial planes. A zero-chirp case checks identity of the implementation, and a measured opposite-sign chirp provides a secondary descriptive comparison. The central inference remains limited to the registered negative-chirp condition and the specified collection geometry.

2 Related Work

Finite-energy Airy beams and their generation and control are treated in the literature [2, 4]. The verified bibliographic record for Zhang's study identifies propagation characteristics of chirped circular Airy beams [8]; Zhan and coauthors' title identifies vortex Airy beams with quadratic phase modulation [7]. These titles establish close topical relevance to phase-modified Airy-beam

propagation. The available verified material here is bibliographic metadata rather than full-text evidence, so it does not support a detailed comparison of their field definitions, phase conventions, collection observables, or numerical conditions with the present calculation.

Pearcey and butterfly fields are other structured beam families studied in propagation contexts [3, 5], but they are not the input fields in this study. Gaussian resonator theory provides relevant background for quadratic phase in optical fields [1]; it does not imply that this Airy source is closed under Gaussian beam-parameter evolution. The present work therefore makes a narrow numerical comparison for a finite-energy Airy source and a specified fixed-axis collection measure. A comprehensive distinction from prior chirped-Airy studies would require direct examination of their full texts and matched definitions, and is not inferred here.

3 Theoretical model

3.1 Finite-energy Airy input and phase intervention

The source profile is

$$V(x,y) = \mathrm{Ai}\Big(\frac{x}{s}\Big)\,\mathrm{Ai}\Big(\frac{y}{s}\Big)\exp\Big[\alpha\frac{x+y}{s}\Big] \tag{1}$$

inside the physical aperture, and zero outside it. Here $s = 0.20\,\mathrm{mm}$ is the Airy scale, $\alpha = 0.10$ is the apodization parameter, and the aperture radius is $R_{\mathrm{ap}} = 0.9\,\mathrm{mm}$. The setup waist is $w = 0.30\,\mathrm{mm}$; it is not the Airy scale.

The executed chirp multiplies this field by

$$V_c(x,y) = V(x,y)\exp\left[ic\frac{x^2+y^2}{w_\phi^2}\right], \tag{2}$$

where $c = -0.5$ for the registered condition and $w_\phi = 0.30\,\mathrm{mm}$ is the executed optical setup waist. Thus the Airy scale s and phase radius w_ϕ are distinct parameters. Because the multiplier has unit modulus, the chirp changes phase while preserving source-plane amplitude.

For sampled values V_{ab}, the normalized input field is

$$U_0(x_j,y_k) = \sqrt{\frac{P}{\Delta x^2\sum_{a,b}|V_{ab}|^2}}\,V(x_j,y_k). \tag{3}$$

Here P is the input power and $\Delta x = \Delta y$ is the pixel spacing. This normalization sets the sampled input power to P; the transformed and base arms are separately normalized to the same input power.

3.2 Propagation and collection observable

The primary propagation model is the angular-spectrum method (ASM):

$$U_z = \mathcal{F}^{-1}\Big[\mathcal{F}(U_0)B_z\exp\Big(i2\pi z\sqrt{\lambda^{-2}-f_x^2-f_y^2}\Big)\Big], \tag{4}$$

where λ is the wavelength, f_x, f_y are transverse spatial frequencies, and B_z is a conservative radial spectral cutoff. For computational-window width L, the cutoff frequency is

$$\nu_c = \min\left[\lambda^{-1}, \frac{\lambda^{-1}}{\sqrt{1+(2z/L)^2}}\right]. \tag{5}$$

This is a conservative radial cutoff and not the complete Matsushima passband construction.

The fraction of input power inside a circular core of radius r_c, centered on the fixed physical grid origin, is

$$C(z) = P^{-1} \int_{x^2+y^2 \le r_c^2} |U_z|^2 \, dA, \qquad \overline{C} = \mathrm{median}_{z \in \mathcal{Z}} \, C(z). \tag{6}$$

The radius is $r_c = 110\,\mu$m. Numerically, the integral is the sum of pixel intensities whose pixel centers satisfy $r \le r_c$, multiplied by Δx^2. The primary registered comparison is

$$\rho = \frac{\overline{C}_{c=-0.5}}{\overline{C}_{\mathrm{base}}}, \tag{7}$$

where each axial median is calculated separately before taking the ratio. The prediction is $\rho \ge 1$. This statistic is a fixed-axis power fraction, not a peak-intensity metric.

The propagation system also supports coherent composition of a field U with a rotated copy:

$$U_c = \sqrt{\frac{P}{\int |U + ae^{i\delta}\mathcal{R}_\theta U|^2 dA}} \left(U + ae^{i\delta}\mathcal{R}_\theta U\right). \tag{8}$$

Here $\mathcal{R}_\theta$ denotes rotation by angle θ, δ is the relative phase, and a is the relative amplitude weight. The prefactor normalizes the composed field to power P; an exactly zero coherent field cannot be normalized. The executed comparison has $\theta = 0$, $\delta = 0$, and $a = 1$, but its measured arms are single-field base and chirped inputs, not a two-arm coherent composition. No rotation is applied. In the general implementation, rotation samples the real and imaginary fields independently by bilinear interpolation with constant zero boundaries about the physical grid origin, then normalizes the rotated arm to input power.

4 Numerical methods

The registered primary calculation uses $N = 384$ samples per transverse dimension on a 3.6 mm computational window, wavelength $\lambda = 633$ nm, input power $P = 1$ W, and the 0.9 mm physical aperture. Coordinates are

$$x_j = (j - \lfloor N/2 \rfloor)L/N, \qquad y_k = (k - \lfloor N/2 \rfloor)L/N.$$

The primary propagator is ASM, with no lens applied. The 11 equally spaced planes from 0.04 to 0.08 m are separated by 4 mm. The executed setup waist is 0.30 mm, which sets the chirp phase radius in Eq. (2). Transformed and base arms share the source parameters, aperture, power, wavelength, propagation model, and axial planes.

The confirmation reference uses $N = 576$, the same 3.6 mm window and physical parameters, and the same 11 axial planes. Additional checks vary transverse mesh, computational window, and axial sampling, and compare representative calculations with an independent paraxial Fresnel propagator. These calculations test numerical stability of the paired comparison; they are not independent physical replicates or an estimate of experimental uncertainty. A numerical error floor of 0.01 is reported for the confirmation.

5 Results and physical interpretation

5.1 Registered negative-chirp comparison

The registered 384×384 calculation gives

$$\overline{C}_{c=-0.5} = 0.072402, \qquad \overline{C}_{\mathrm{base}} = 0.062833, \qquad \rho = 1.1528.$$

Table 1: Fixed-axis core fractions and transformed-to-base ratios. The primary and reference rows use the same registered source, chirp, physical aperture, wavelength, core, and axial interval. The +0.5 row is a secondary, measured opposite-sign comparison.

Condition and numerical tier	$\overline{C}_{\text{transformed}}$	$\overline{C}_{\text{base}}$	ρ
$c = -0.5$, primary 384×384	0.072402	0.062833	1.1528
$c = -0.5$, reference 576×576	0.071592	0.062104	1.1528
$c = 0$, identity control	0.062104	0.062104	1.0000
$c = +0.5$, reference 576×576	0.054412	0.062104	0.8761

The 576×576 reference gives

$$\overline{C}_{c=-0.5} = 0.071592, \qquad \overline{C}_{\text{base}} = 0.062104, \qquad \rho = 1.1528.$$

Both ratios exceed unity. The reference result exceeds the reported numerical error floor by a margin of approximately 0.143 in the ratio, supporting the registered directional prediction for this numerical regime.

The ratio is stable across the supplied numerical checks. The recorded mesh, expanded-window, denser-axial-grid, and independent Fresnel comparisons give ratios of approximately 1.1528, 1.1528, 1.1528, and 1.1528, respectively. The quantization check gives 1.1531, and the held-out wavelength check gives 1.1540. These are repeat calculations and perturbation checks of the same model comparison, not independent physical observations. They support stability of the paired ratio under the tested numerical variations; they do not imply that absolute core fractions are invariant to discretization or window choice.

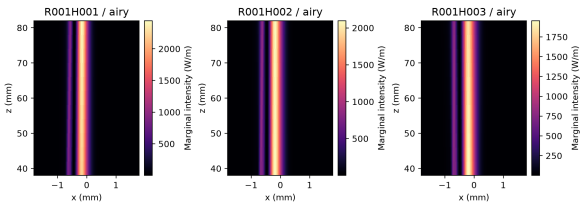
The zero-chirp control is an identity control: the transformed and base fields coincide, with a confirmed ratio of 1.0 in the primary wavelength condition. Its role is to verify implementation, not to supply a nonzero physical effect.

5.2 Opposite-sign comparison and alignment diagnostic

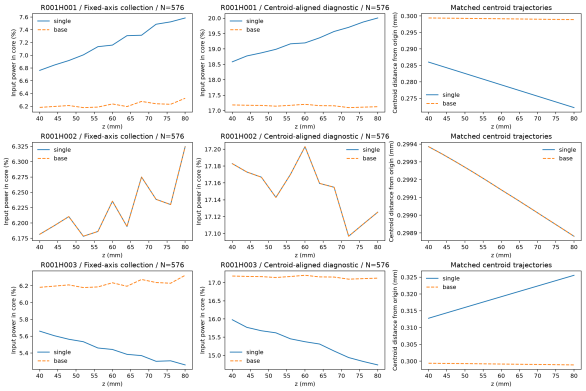
The +0.5 reference calculation yields a fixed-axis median of 0.054412, compared with 0.062104 for its base arm, giving $\rho = 0.8761$. Unlike the negative-chirp condition, the measured positive-chirp effect is below unity. The opposite-sign condition is a secondary comparison and does not change the frozen primary prediction. The two signed values show different outcomes at these two tested settings; they do not establish a continuous chirp response or a general sign law.

For the $c = -0.5$ reference condition, centroid-aligned core efficiencies are approximately 0.1920 for the chirped arm and 0.1716 for the base arm, compared with fixed-axis medians of 0.0716 and 0.0621. Recentring the observation circle therefore changes collection substantially for both arms. Plane-by-plane alignment diagnostics give fixed-axis differences of about 0.0058–0.0129 across the sampled planes, while centroid-centered differences are about 0.0140–0.0288. The difference between these measures varies along the interval. This is descriptive evidence that the fixed-axis endpoint is sensitive to alignment as well as to the beam-centered distribution; it is not a causal decomposition of the effect into displacement and deformation.

A phase-gradient change is a physically plausible route by which the chirp can affect propagated collection: Eq. (2) leaves input intensity unchanged but changes the phase slope that enters propagation. The measurements establish the change in the chosen collection statistic, not that this proposed route is the unique cause. A matched analysis designed to distinguish displacement from changes in the beam-centered distribution would be needed for that inference.



(a) Propagation view of the recorded comparison.



(b) Matched axial diagnostics.

Figure 1: Complementary views of the negative-chirp and base Airy calculations. The axial diagnostics relate fixed-axis collection to alignment-centered collection; they do not identify the cause of their difference.

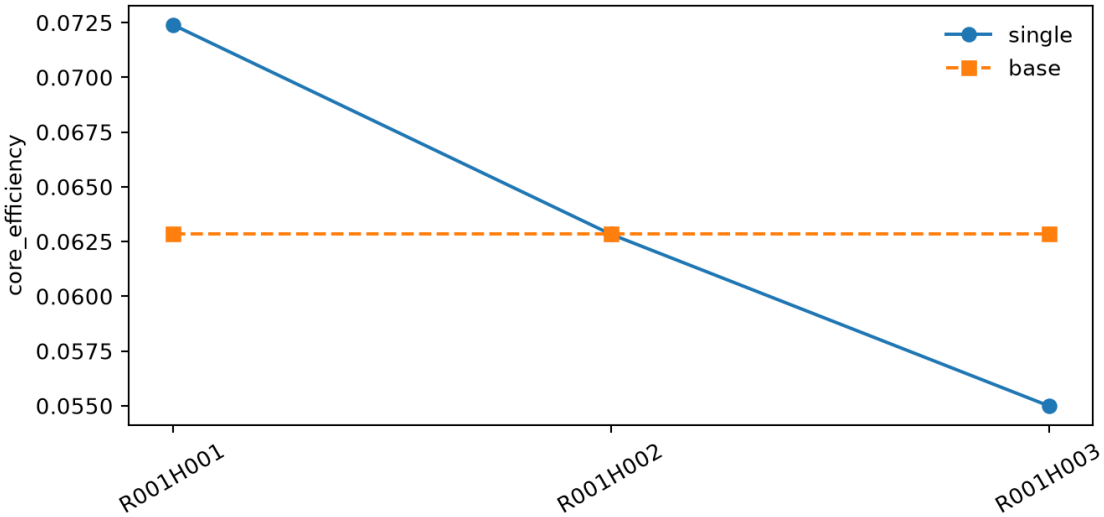


Figure 2: Recorded research-result overview. The registered inference is based on the ratio of the separately computed axial medians of fixed-axis core efficiency, not on visual inspection of this summary.

5.3 Context of absolute collection controls

Separate optimizations of classical phase controls report absolute core efficiencies of 0.3087 for defocus, 0.3351 for a quadratic–quartic phase, and 0.3402 for a four-parameter phase. These are optimized absolute values for different phase settings, not matched transformed-to-base ratios for the registered Airy comparison. They are not a controlled superiority benchmark and do not convert the relative increase in Eq. (7) into a claim of superior absolute collection.

5.4 Figures

Figures 1a and 1b are grouped because they show complementary views of the same chirped-versus-base propagation comparison: the first provides spatial and axial context, and the second presents matched axial diagnostics, including the fixed-axis and alignment-centered measures relevant to interpreting the endpoint. Figure 2 is a separate overview; none of the inferences is drawn from visual appearance alone.

6 Conclusions

For the specified finite-energy Airy source, the executed quadratic phase with $c = -0.5$ produces a transformed-to-base median fixed-axis core-efficiency ratio of 1.1528 in both the primary calculation and the reference calculation to four decimal places. The supplied mesh, window, axial-sampling, and independent Fresnel checks support numerical stability of the paired ratio, and the zero-chirp identity control is validated. The measured $+0.5$ comparison is below unity, but does not establish a general chirp-sign response.

The result concerns the fixed-axis fraction of power collected in a 110 μm radius core over 0.04–0.08 m, at 633 nm, for the stated source and aperture. The higher centroid-aligned values show why collection must be interpreted with its alignment convention specified; they do not isolate a unique mechanism. The available evidence supports the registered numerical comparison, but not a broader physical or prior-art conclusion.

Acknowledgment

Auto BeamScientist adapts Sakana AI's AI Scientist-v2 framework [6].

References

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